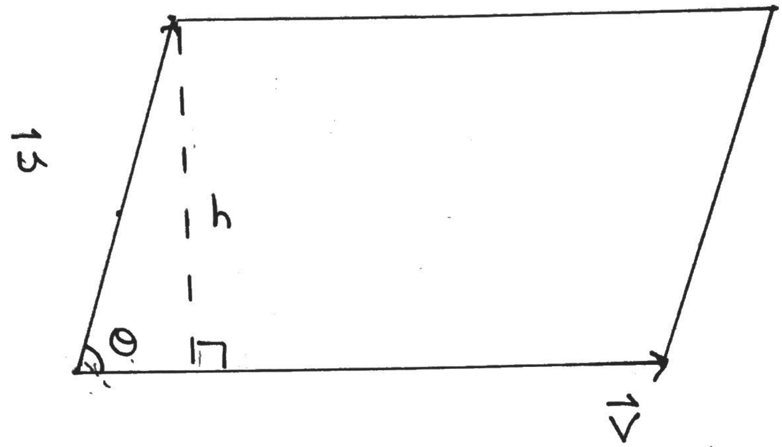
Area of a Parallelogram



$$\text{Area} = (\text{Base})(\text{Height})$$

$* \sin(\theta) = \dfrac{h}{|\vec{u}|}$

$$\text{Area} = |\vec{v}| \cdot |\vec{u}| \sin(\theta)$$

$h = |\vec{u}| \sin(\theta)$

$$\text{Area} = |\vec{u}| \cdot |\vec{v}| \sin(\theta)$$

$$\text{Area} = |\vec{u} \times \vec{v}|$$

$$\therefore \text{Area} = |\vec{u} \times \vec{v}|$$

Example: Determine the area of a triangle with vertices $P(7, 2, -5)$, $Q(9, -1, -6)$ and $R(7, 3, -3)$

Solution

* find the vectors that form 2 sides of the triangle

$\vec{PQ} = [2, -3, -1]$ and $\vec{PR} = [0, 1, 2]$

$\vec{PQ} \times \vec{PR}$:

$$\begin{array}{cccccc} 2 & -3 & -1 & 2 & -3 & -1 \\ & & \times & \times & \times & \\ 0 & 1 & 2 & 0 & 1 & 2 \end{array}$$

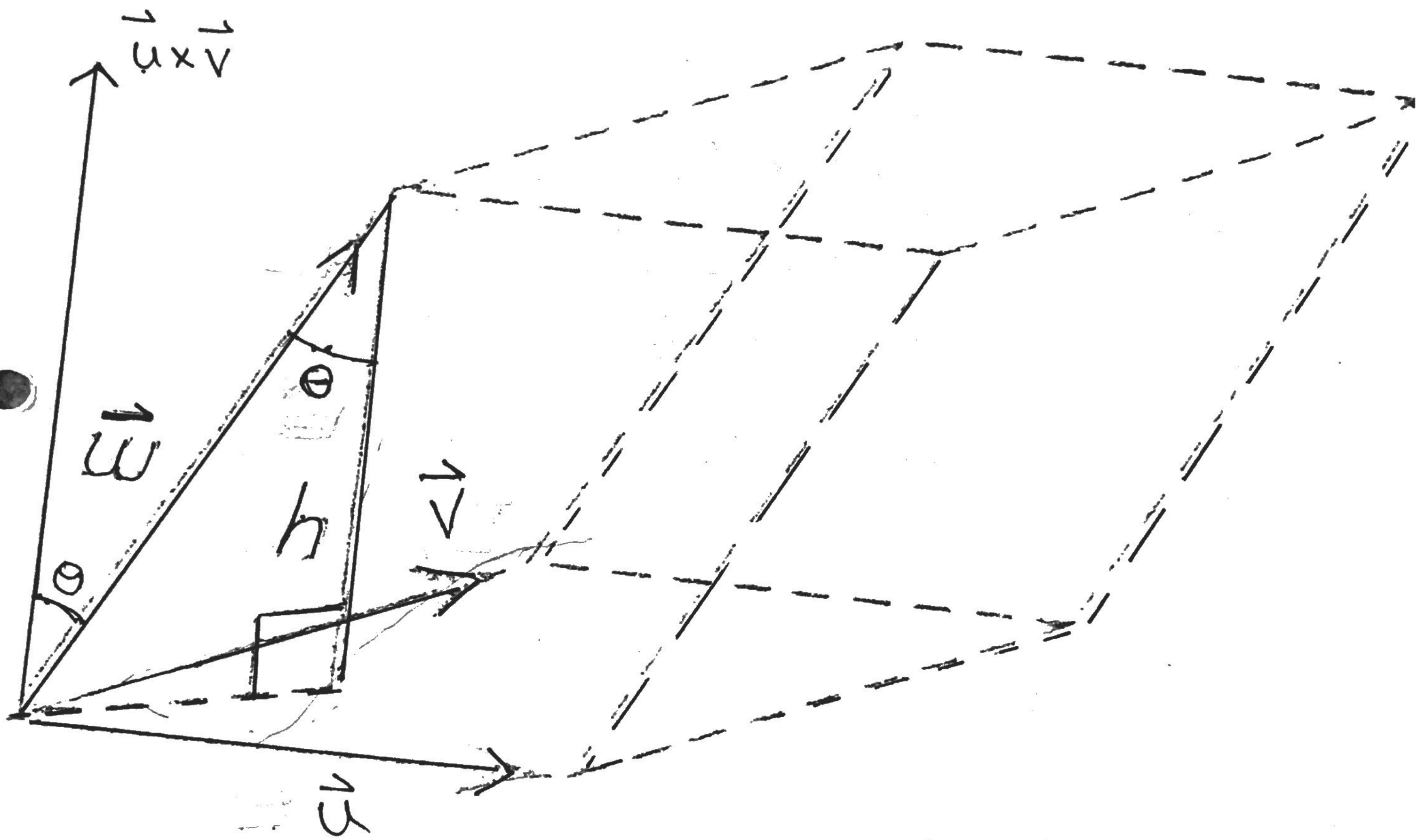
$\vec{PQ} \times \vec{PR} = [-5, -4, 2]$

$|\vec{PQ} \times \vec{PR}| = \sqrt{25 + 16 + 4}$

$|\vec{PQ} \times \vec{PR}| = \sqrt{45}$ ← area of parallelogram

$\therefore$ Area of Triangle $= \frac{1}{2} \cdot \sqrt{45}$ units^2

Parallelepiped



* $\cos(\theta) = \dfrac{h}{|\vec{w}|}$

* $h = |\vec{w}|\cos(\theta)$

Volume of a Parallelepiped

$$V = (\text{Area of Base}) \cdot (\text{Height})$$

$$V = \left(\overset{|\vec{A}|}{|\vec{u} \times \vec{v}|}\right)\left(\overset{|\vec{B}|}{|\vec{w}| \cos(\theta)}\right)$$

$$\boxed{V = \left|(\vec{u} \times \vec{v}) \cdot \vec{w}\right|}$$

absolute value.

* If $(\vec{u} \times \vec{v}) \cdot \vec{w}$ is negative, simply take the absolute value to obtain the volume.

Example: Determine the volume of a parallelepiped determined by the vectors $\vec{u} = [2, -5, -1]$, $\vec{v} = [4, 0, 1]$ and $\vec{w} = [3, -1, -1]$

Solution:

$$V = (\vec{u} \times \vec{v}) \bullet \vec{w}$$

$$V = [-5, -6, 20] \bullet [3, -1, -1]$$

$$\begin{matrix} 2 & -5 & -1 & 2 & -5 & -1 \\ 4 & 0 & 1 & 4 & 0 & 1 \end{matrix}$$

$$= [-5, -6, 20]$$

$$V = -15 + 6 - 20$$

$$V = |-29|$$

Take absolute value

$$\therefore V = 29 \text{ units}^3$$